

ASSIGNMENT - 2

Fourier Series of Even and Odd functions

1. Obtain the Fourier Series Expansion of $f(x) = |\cos x|$ $(-\pi, \pi)$

$$\text{Ans: } a_0 = \frac{4}{\pi} \quad a_n = -\frac{4 \cos(n\pi/2)}{\pi(n^2-1)}, \quad b_n = 0$$

2. Find the Fourier series of $f(x) = |x|$ in $(-1, 1)$

$$\text{Ans: } a_0 = 1, \quad a_n = \frac{-2}{n^2\pi^2} \{1 - (-1)^n\}, \quad f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{-2}{n^2\pi^2} \{1 - (-1)^n\} \cos n\pi x$$

3. Find the Fourier series of $f(x) = \begin{cases} 0 & -2 < x < -1 \\ 1+x & -1 < x < 0 \\ 1-x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$

$$\text{Ans: } \frac{1}{4} + \frac{4}{n^2} \sum_1^{\infty} \frac{1}{n^2} \left[1 - \cos \frac{n\pi}{2} \right] \cos \frac{n\pi x}{2}$$